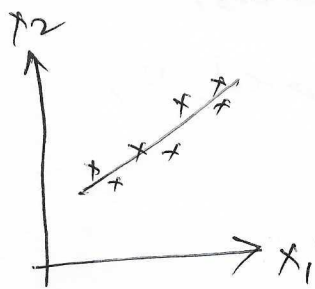


Principal Component Analysis (PCA).

Dimensionality reduction algorithm.



Often $\vec{x}$ may lie in a high dimensional space, but really occupies a small "lower dimension" subspace.

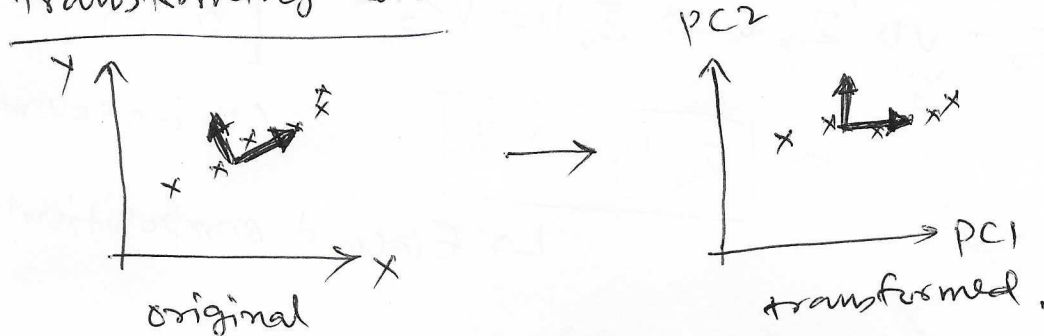
Goal: $\vec{x} \in \mathbb{R}^d \rightarrow \vec{z} \in \mathbb{R}^k$
 $k < d$.

Different views of PCA:

- Maximum variance view
- Minimum reconstruction error view
- KL transform view (we'll see this)

} Ultimately all views lead to the same interpretation.

"Transforming" data.



Let $\vec{x} \in \mathbb{R}^d \rightarrow \vec{z} \in \mathbb{R}^d$ (first transform, reduce dimension later)

$\vec{z} = U^T \vec{x}$ (U is orthonormal. why?)

Assumption: $E[\vec{x}] = 0$

If not, $\vec{x} = \vec{x} - E[\vec{x}]$ (centering the data)

$$\Sigma_x = E[(\vec{x} - E[\vec{x}])(\vec{x} - E[\vec{x}])^T]$$

$$= E[\vec{x} \vec{x}^T]$$

$$= \begin{bmatrix} \ddots & E[x_i x_j] \\ & \ddots \\ E[x_i^2] & \end{bmatrix}_{d \times d}$$

$$\text{cov}(x, y) = E[(x - E[x])(y - E[y])]$$

↓
measures joint variability of two random variables.

Goal: $Z = U^T X$

$$U_i^T U_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \quad (\text{basis vectors})$$

Find U such that $\vec{z} \in \mathbb{R}^d$ are un-correlated, i.e.,

how? $\Sigma_z = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix}_{d \times d}$

$$\begin{aligned} \Sigma_z &= E[(z - E[z])(z - E[z])^T] \\ &= E[zz^T] \quad (\text{since } E[X] = 0, \text{ \& it's a linear transform}) \\ &= E[U^T X (U^T X)^T] = E[U^T X X^T U] = U^T E[XX^T] U \end{aligned}$$

$$\Rightarrow \Sigma_z = U^T \Sigma_X U$$

$$\Rightarrow U \Sigma_z = \underbrace{U U^T}_I \Sigma_X U \Rightarrow \Sigma_X U = U \Sigma_z = U \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix}$$

$$\Rightarrow \boxed{\Sigma_X U_i = \lambda_i U_i}$$

(U_i is a column of U)

$\hookrightarrow$ Eigen decomposition!

- Start with $N \times d$ data matrix X
- Recenter: $X_c = X - \bar{X}$
- Compute Covariance matrix $\Sigma = \frac{1}{N} X_c^T X_c$
- Find eigen vectors & eigen values of Σ .
- Principal Components: k eigen-vectors with highest eigen-values.

$\nwarrow$ dimensionality reduction!

Each eigenvector represents a principal component direction — a new axis in the feature space. Each eigenvalue tells how much variance is captured along corresponding eigenvector.

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Scaling up.

- Σ is $d \times d$
- large no. of features
- finding eigenvalues is very slow.

Consider an image 100×100

so, $\vec{x}_i \in \mathbb{R}^{10000}$

$$X = \begin{bmatrix} & & & \end{bmatrix}_{N \times d} \leftarrow d = 10,000$$

So PCA requires $\text{eig}(\frac{1}{N} X^T X) \Rightarrow O(d^3)$

Can we speed up?

↓ Yes, use Singular Value Decomposition (SVD)

$$X = U S V^T$$

orthonormal diagonal orthonormal

$$\begin{aligned}
 X^T X &= (U S V^T)^T U S V^T \\
 &= V S^T \underbrace{U^T U}_I S V^T \\
 &= V S^T S V^T \\
 &= V S^2 V^T
 \end{aligned}$$

SVD can be done in $O(N^2 d)$ time.

This is an eigenvalue system:

$$\begin{aligned}
 X^T X &= V S^2 V^T \\
 \Rightarrow V^T (X^T X) &= S^2 V^T
 \end{aligned}$$

$$\downarrow$$

$$\boxed{A x = \lambda x}.$$

$S_{ii} \rightarrow$ eigenvalue of $X^T X$
 $V^T \rightarrow$ eigenvectors of $X^T X$.

- Start from $N \times d$ data matrix
- $X_c = X - \bar{X}$
- Call SVD on $X_c \rightarrow$ ask for k singular vectors.
- Principal components: k singular vectors with highest singular values